



IBM Developer
SKILLS NETWORK

Predicting Rocket First Stage Reuse with data science

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Outline

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 - EDA, SQL
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Executive Summary

Summary of methodologies

The project aims to identify the key factors that contribute to a successful rocket landing. To achieve this objective, the following methodologies were employed:

- Data Collection: SpaceX REST API & web scraping
- Data Preparation: Created success/failure landing outcome
- Exploratory Analysis: Examined payload, launch site, flight number, and yearly trends
SQL Analysis: Calculated payload statistics and landing success/failure counts
- Geospatial Analysis: Evaluated launch site success rates and nearby geographical features
- Visualization: Identified top-performing launch sites and successful payload ranges
- Machine Learning: Predicted landing outcomes using: Logistic Regression, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN)

Summary of all results

Exploratory Data Analysis (EDA)

- Launch success rate increased over time.
- KSC LC-39A achieved the highest landing success rate.
- ES-L1, GEO, HEO, and SSO orbits recorded 100% landing success.

Visualization

Most launch sites are located in proximity to the equator, and close to the coast.

Introduction

- Project background and context
 - SpaceX reduces launch costs by reusing Falcon 9 first-stage boosters.
 - Falcon 9 launches cost ~\$62M, compared to \$165M+ for non-reusable rockets.
 - Predicting first-stage landing success helps estimate launch costs and reusability.
 - This project uses public SpaceX data and machine learning to predict landing outcomes.
- Problems you want to find answers
 - Predict whether the Falcon 9 first-stage booster will land successfully using machine learning.
- GitHub URL of the project: <https://github.com/evlis-i/IBM-Applied-DS>

Section 1

Methodology



Methodology

Executive Summary

- **Collect Data:** SpaceX REST API & web scraping
- **Prepare Data:** Clean, filter, handle missing values, and apply one-hot encoding
- **Analyze Data:** Perform EDA using SQL and data visualizations
- **Visualize Results:** Create interactive maps and dashboards with **Folium** and **Plotly Dash**
- **Build Models:** Train, tune, and evaluate classification models to predict landing success

Data Collection

The data on SpaceX launches was supposed to be obtained from the SpaceX REST API, which provides detailed information in JSON format. The dataset includes information about the rocket used, payload specifications, launch details, landing specifications, and landing outcomes. However, due to a communication issue with the SpaceX REST API, the data could not be retrieved successfully. Requests to endpoints under `api.spacexdata.com/v4/...` consistently returned an “**SSL handshake failed – Error code 525**” response, preventing access to the required data.

In addition to the REST API, Falcon 9 launch data was also collected through web scraping from Wikipedia using the BeautifulSoup library. The relevant launch information was extracted from Wikipedia tables and saved as a CSV file for further analysis.

Data Collection – SpaceX API

- Requests to endpoints under `api.spacexdata.com/v4/...` consistently returned an “**SSL handshake failed – Error code 525**” response, preventing access to the required data.
- HTTP 525 is a Cloudflare error meaning the origin server (`api.spacexdata.com`) failed to complete an SSL handshake with Cloudflare's proxy.
- This is a server-side SSL/TLS issue with the SpaceX data API.
- It's an infrastructure issue on the API's end, not something in the request configuration.

Notebook link in GitHub:

<https://github.com/evlis-i/IBM-Applied-DS/blob/main/Modul1-spacex-data-collection-api.ipynb>

Data Collection – Web Scraping

To obtain Falcon 9 launch records, web scraping was performed using the BeautifulSoup library. The launch records were extracted from an HTML table on Wikipedia containing information about Falcon 9 launches. The HTML table was parsed using BeautifulSoup and the relevant rows and columns were extracted. The collected data was added to dictionary then converted into a Pandas DataFrame, allowing it to be easily cleaned, processed, and analyzed using Python.

GitHub URL of the completed web scraping notebook:

<https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul1-spacex-webscraping.ipynb>

Steps:

- Request data (Falcon 9 launch data) from Wikipedia
- Create BeautifulSoup object from HTML response
- Extract column names from HTML table header
- Collect data from parsing HTML tables
- Create dictionary from the data
- Create dataframe from the dictionary
- Export data to csv file

Data Wrangling

- Perform exploratory Data Analysis and determine Training Labels
- Calculate:
 - the number of launches on each site
 - the number and occurrence of each orbit
 - the number and occurrence of mission outcome of the orbits
- Create a landing outcome label from Outcome column
 - RTLS (Return To Launch Site) True to 1 and False to 0
 - True Ocean to 1 and False Ocean to 0
 - ASDS (Autonomous Spaceport Drone Ship) True to 1 and False to 0
 - None to 0
- Export data to csv file
- GitHub URL: <https://github.com/evlis-i/IBM-Applied-DS/blob/main/Modul1-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

- Scatter Plots

- FlightNumber vs. PayloadMass
- FlightNumber vs LaunchSite
- Payload Mass vs. Launch Site
- FlightNumber vs. Orbit type
- Payload Mass vs. Orbit

- Bar Plots

- Success Rate vs. Orbit Type

- Line Charts

- Success Rate vs Year

- GitHub URL: <https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul2-spacex-eda-datavisualization.ipynb>

EDA with SQL

Queries to display or list information about SpaceX data

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
- List the records which will display the month names, failure landing outcomes in drone ship ,booster versions, launch site for the months in year 2015.
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

GitHub URL: <https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul2-spacex-EDA-SQL.ipynb>

Build an Interactive Map with Folium

- Mark all launch sites on a map
 - From a csv dataset we created a DataFrame with launch site , latitude longitude
 - We created a folium map
 - We create and add folium.Circle and folium.Marker for each launch site on the site map
- Mark the success/failed launches for each site on the map
 - We assigned the dataframe launch outcomes(failures, successes) to classes 0 and 1 with Green and Red markers on the map in a MarkerCluster()
- Calculate the distances between a launch site to its proximities
 - Draw lines that show the distance from a launch site to coast line or other proximities
 - Create a launch site marker with distances to a closest city, railway, highway, Calculate the distances.
- GitHub URL : <https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul3-spacex-folium.ipynb>

Build a Dashboard with Plotly Dash

The dashboard contains 2 parts:

- The first part shows the total launches by a certain site/all sites.
 - It contains a Launch Site Drop-down Input Component
 - A pie-chart to display relative proportions
 - A callback function renders success-pie-chart based on selected site dropdown.
- The second shows the relationship with Outcome and Payload Mass (Kg) for the different Booster Versions
 - It has a Range Slider to Select Payload
 - A scatter plot to display the success/failed launches
 - A callback function renders the success-payload-scatter-chart scatter plot
- GitHub URL: https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul3_spacex-dash-app.py

Predictive Analysis (Classification)

- The processed data was loaded and prepared using NumPy and Pandas, followed by the necessary data transformations and the division of the dataset into training and testing sets.
- Several machine learning classification models were then developed and evaluated.
- To optimize model performance, different hyperparameter combinations were tested using GridSearchCV.
- Model accuracy was used as the primary evaluation metric.
- After completing the optimization and evaluation process, the performance of the different models was compared, and the best-performing classification model was selected based on its accuracy.
- GitHub URL: https://github.com/evis-i/IBM-Applied-DS/blob/main/Modul4_SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

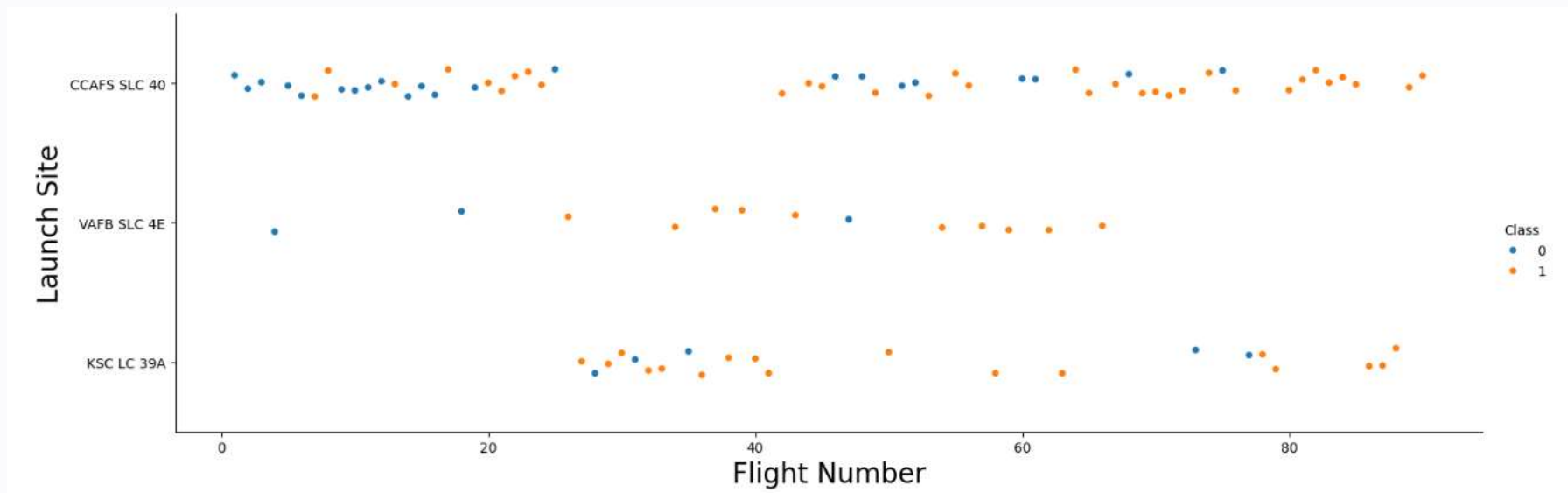
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

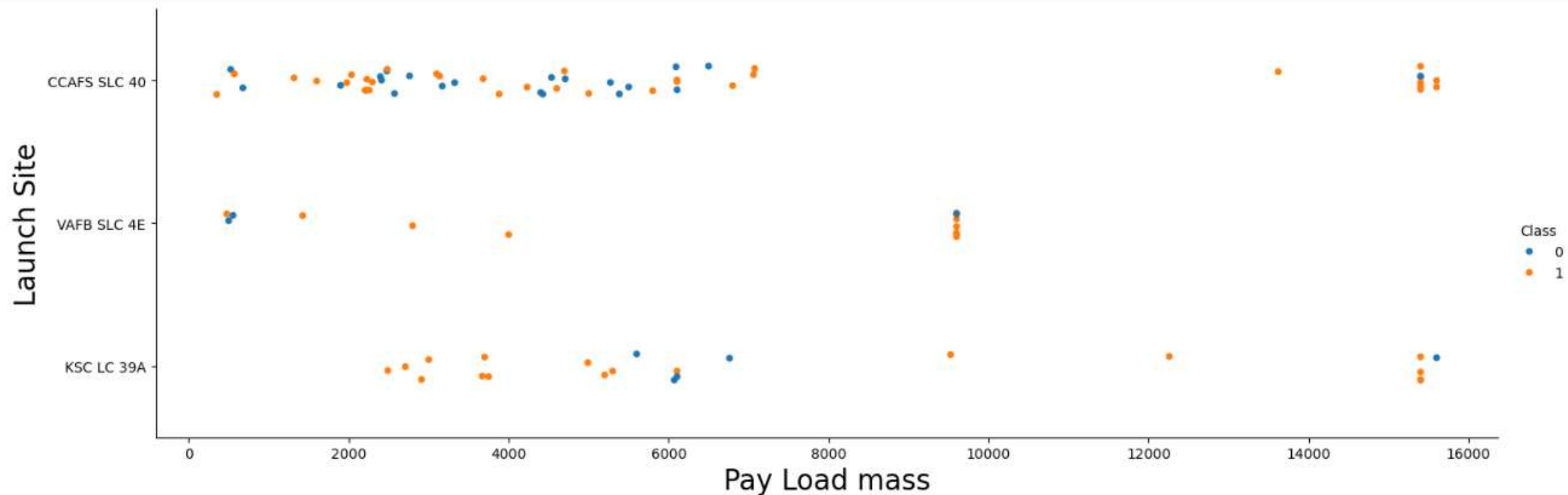


Flight Number vs. Launch Site



The greater the Flight number for each site, the greater the success rate.

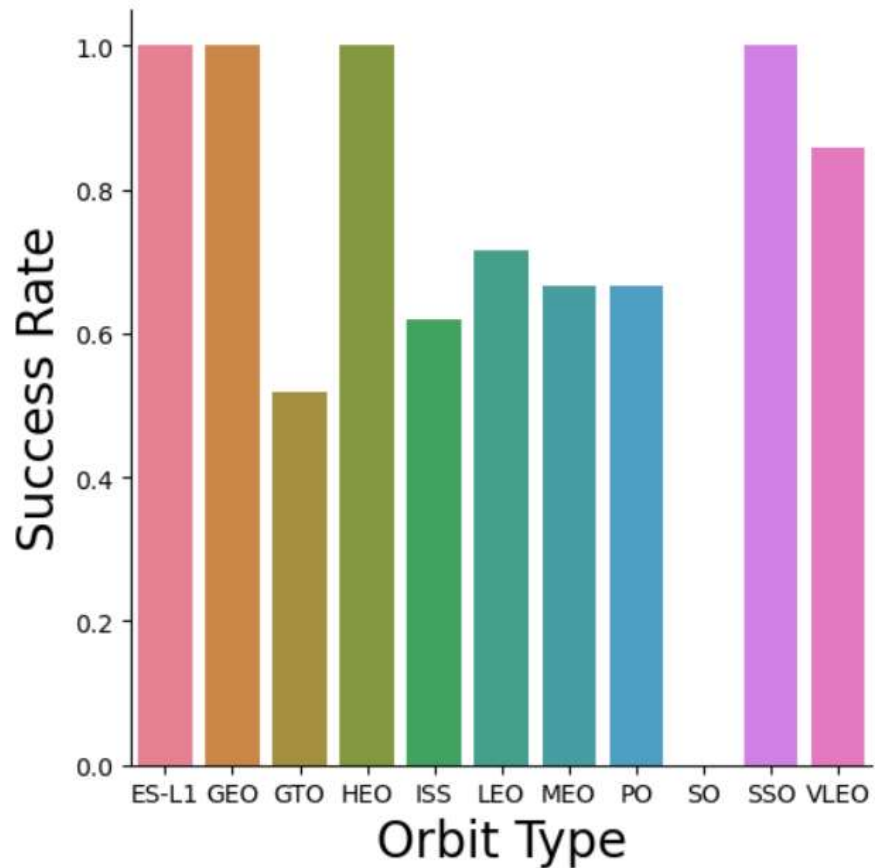
Payload vs. Launch Site



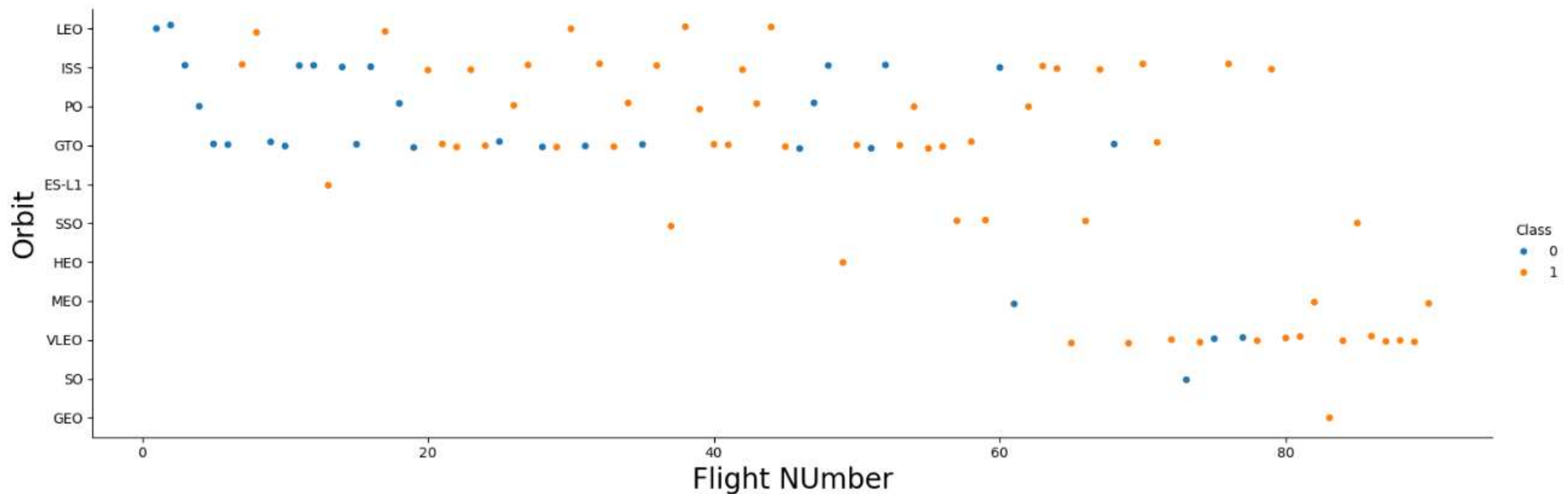
- The greater the payload mass for Launch Site CCAFS SLC 40 the higher the success rate for the Rocket.
- There is not quite a clear pattern to be found using this visualization to make a decision if the Launch Site is dependent on Pay Load Mass for a success launch.

Success Rate vs. Orbit Type

- Orbit Es-L1, GEO, HEO, SSO have the highest success rate
- Orbit SO has 0 success rate, it was only 1 launch to this orbit

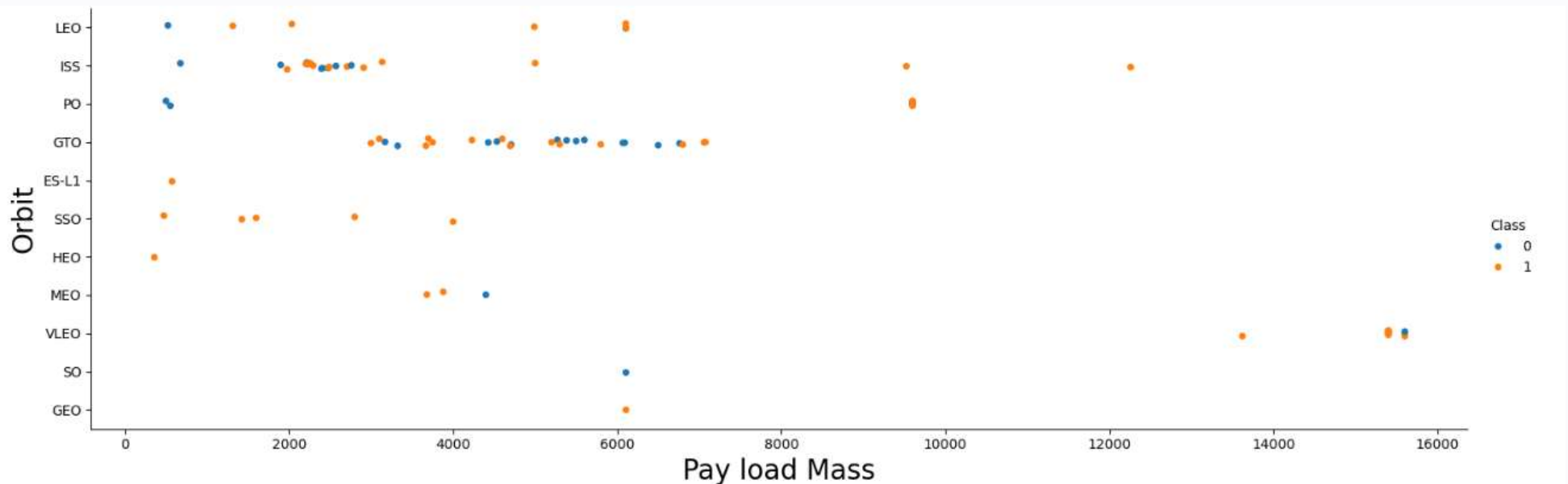


Flight Number vs. Orbit Type



- For some orbits the greater the flight number the greater the success rate like LEO, VLEO.
- For other GTO or PO there's no relationship.

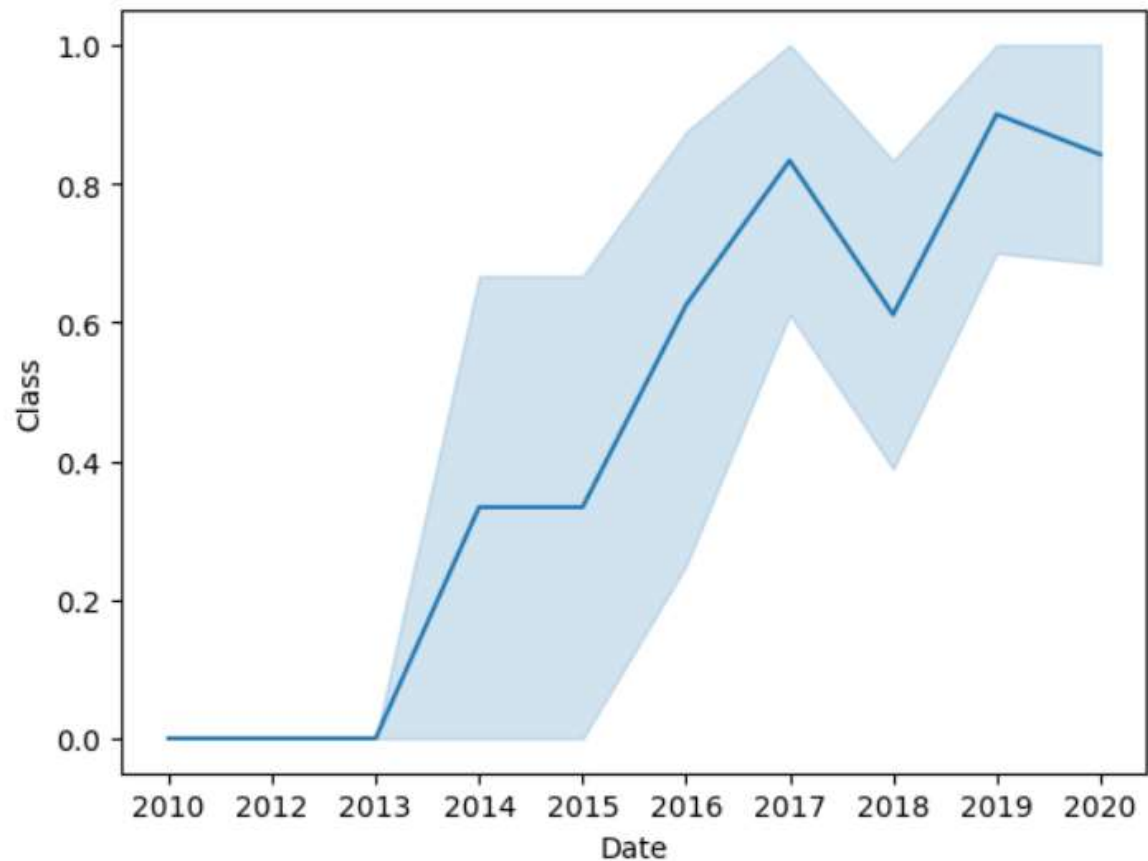
Payload vs. Orbit Type



- It can be observed that for payloads with a heavy weight, the probability of a successful landing is higher for orbits such as PO, LEO, and ISS
- This is not true for the GTO

Launch Success Yearly Trend

- The success rate keeps increasing over time



All Launch Site Names

Using the word distinct in the query means that it will only show Unique values in the Launch_Site column from SPACEXTABLE

Display the names of the unique launch sites in the space mission

```
%sql select distinct Launch_Site from SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEXTABLE where Launch_Site like "CCA%" limit 5
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Using the words “**limit 5**” in the query displays only 5 records. ‘CCA%’ the percentage in the end suggests that the Launch_Site name must start with CCA.

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql select SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE where Customer = "NASA (CRS)"
```

```
* sqlite:///my_data1.db
```

Done.

<u>SUM(PAYLOAD_MASS_KG_)</u>

45596

The total payload carried by boosters from NASA is calculated by using SUM() function on Payload_mass_kg column and filtering the customer.

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS_KG_) FROM SPACEXTABLE where Booster_Version like "F9 v1.1%"
```

```
* sqlite:///my_data1.db  
Done.
```

avg(PAYLOAD_MASS_KG_)

2534.6666666666665

The average payload mass carried by booster version F9 v1.1 is calculated by using the avg() function on PAYLOAD_MASS_KG column and filtering on the Booster_version column.

First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql select min(DATE), Landing_Outcome from SPACEXTABLE where Landing_Outcome like "Success (ground pad)"
```

```
* sqlite:///my_data1.db
```

Done.

min(DATE)	Landing_Outcome
-----------	-----------------

2015-12-22	Success (ground pad)
------------	----------------------

The date of the first successful landing outcome on ground pad is listed by using the min() function on Date Column and filtering on Landing Outcome column

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- *%sql select Booster_Version from SPACEXTABLE where Landing_Outcome like "Success (drone ship)" and 4000<PAYLOAD_MASS_KG_<6000*

Selecting only Booster_Version the the table

The WHERE clause filters the dataset to Landing_Outcome = Success (drone ship)

The AND clause specifies additional filter conditions

Payload_MASS_KG_ between 4000 and 6000

Booster_Version
F9 FT B1021.1
F9 FT B1022
F9 FT B1023.1
F9 FT B1026
F9 FT B1029.1
F9 FT B1021.2
F9 FT B1029.2
F9 FT B1036.1
F9 FT B1038.1
F9 B4 B1041.1
F9 FT B1031.2
F9 B4 B1042.1
F9 B4 B1045.1
F9 B5 B1046.1

Total Number of Successful and Failure Mission Outcomes

Successfull Mission Outcome	Failure Mission Outcome
100	1

- Calculate the total number of successful and failure mission outcomes
 - *%sql select (select count(Mission_Outcome) from SPACEXTABLE where Mission_Outcome like "%Success%") as "Successfull Mission Outcome", (select count(Mission_Outcome) from SPACEXTABLE where Mission_Outcome like "%Failure%") as "Failure Mission Outcome"*
- The query calculates how many SpaceX missions were successful and how many were failures by searching the Mission_Outcome column for the words "Success" and "Failure".
- In the Mission_Outcome column, there were three different strings representing successful mission outcomes, and all of them contained the word "Success." Therefore, to count all successful missions regardless of the specific string, we used the LIKE "%Success%" condition to filter the results.
- First subquery – Successful missions: *SELECT COUNT(Mission_Outcome) FROM SPACEXTABLE WHERE Mission_Outcome LIKE "%Success%"*
- Second subquery – Failed missions: *SELECT COUNT(Mission_Outcome) FROM SPACEXTABLE WHERE Mission_Outcome LIKE "%Failure%"*
- Column aliases: *as "Successfull Mission Outcome" or "Failure Mission Outcome"*

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
 - *%sql select distinct booster_version from SPACEXTABLE where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE)*
- Find the maximum by using the max() function in subquery.
- Select distinct booster versions where PAYLOAD_MASS__KG equals the result of the subquery

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
 - *%sql SELECT CASE SUBSTR(Date, 6, 2) WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March' WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June' WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September' WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December' END AS month_name, Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTABLE WHERE Landing_Outcome like 'Failure (drone ship)' AND SUBSTR(Date, 1, 4) = '2015';*
- We used substr(Date, 6,2) to get the months and substr(Date,0,5)='2015' for years.
- To match month number with month names we used Case When Then function.
- We filtered on landing_outcomes column and date based on year = substr(Date,0,5)

month_name	Booster_Version	Launch_Site	Landing_Outcome
January	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
April	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
 - %sql SELECT Landing_Outcome, count(*) as count_outcomes FROM SPACEXTBL WHERE DATE between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by count_outcomes DESC;*
- Select landing_outcome, filter date by using between, count by using count() function, sort the results in descending order.

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

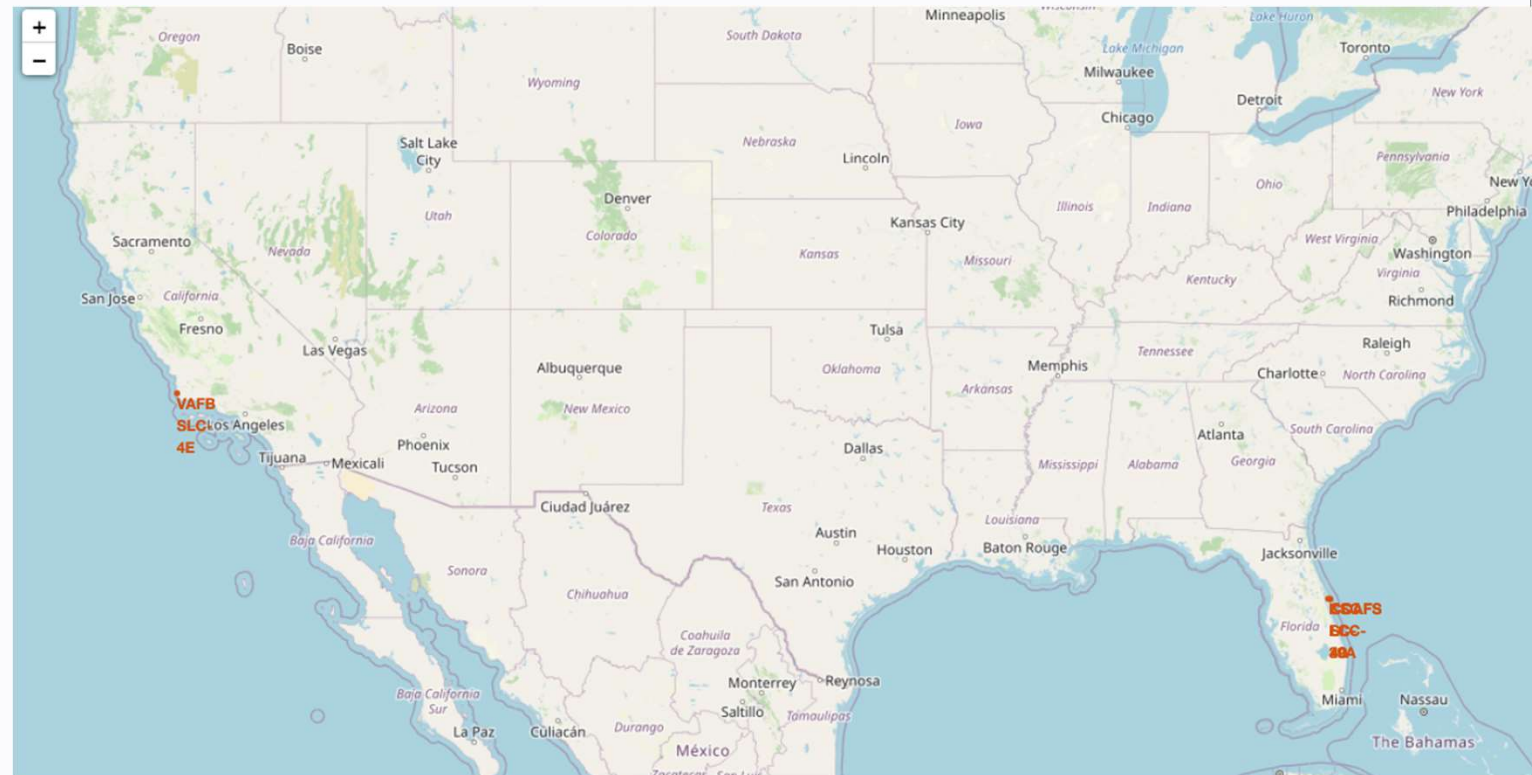
Section 3

Interactive map using Folium



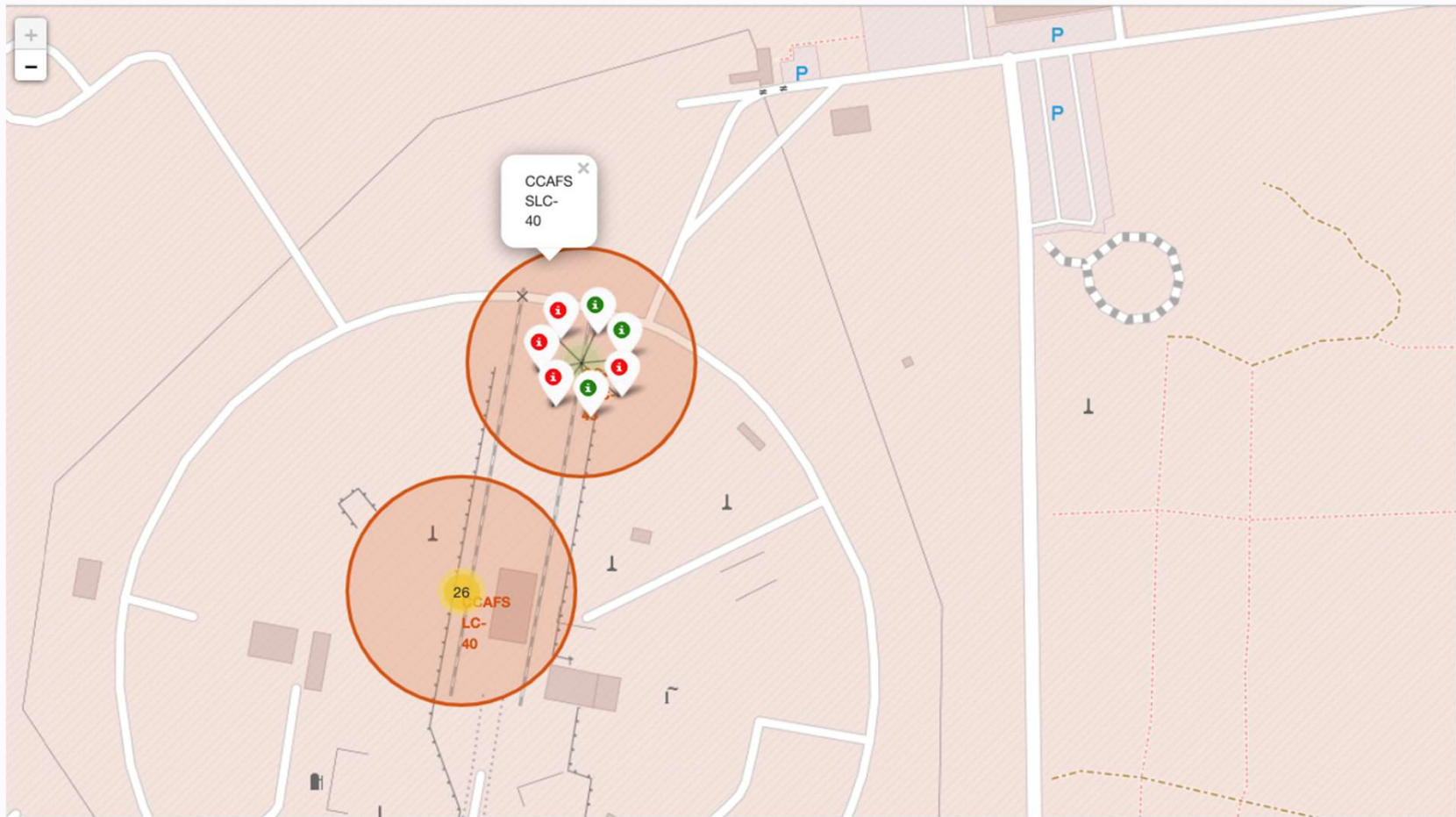
Mark all launch sites on a Folium map

- SpaceX launch sites are in the USA coasts
- They are in south



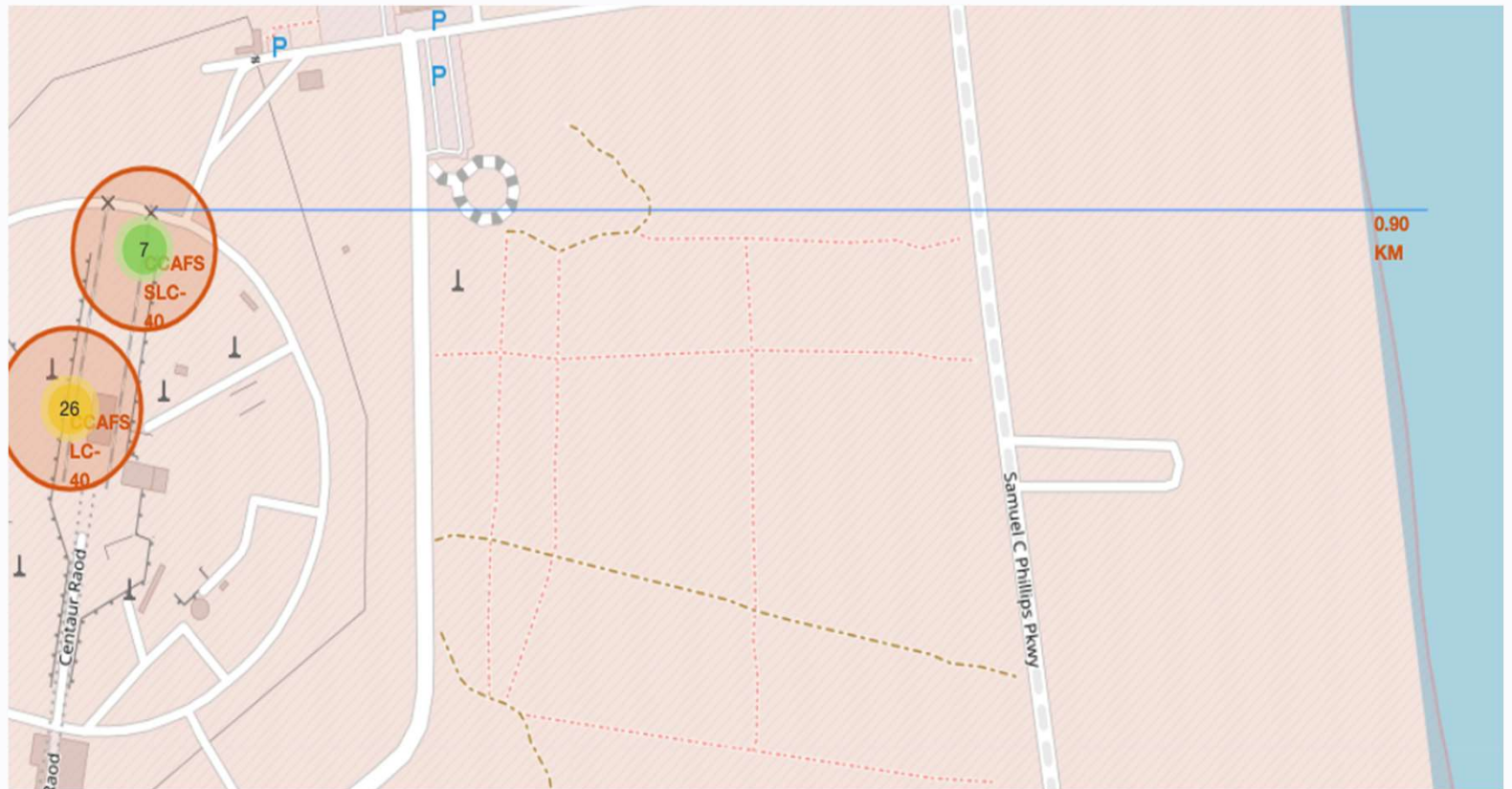
Mark the success/failed launches for each site

- Red is Failure, Green is Success
- For each site if we zoom we can count the success or failure launches



Calculate the distances between a launch site to its proximities

- The distance from a launch site to the coast



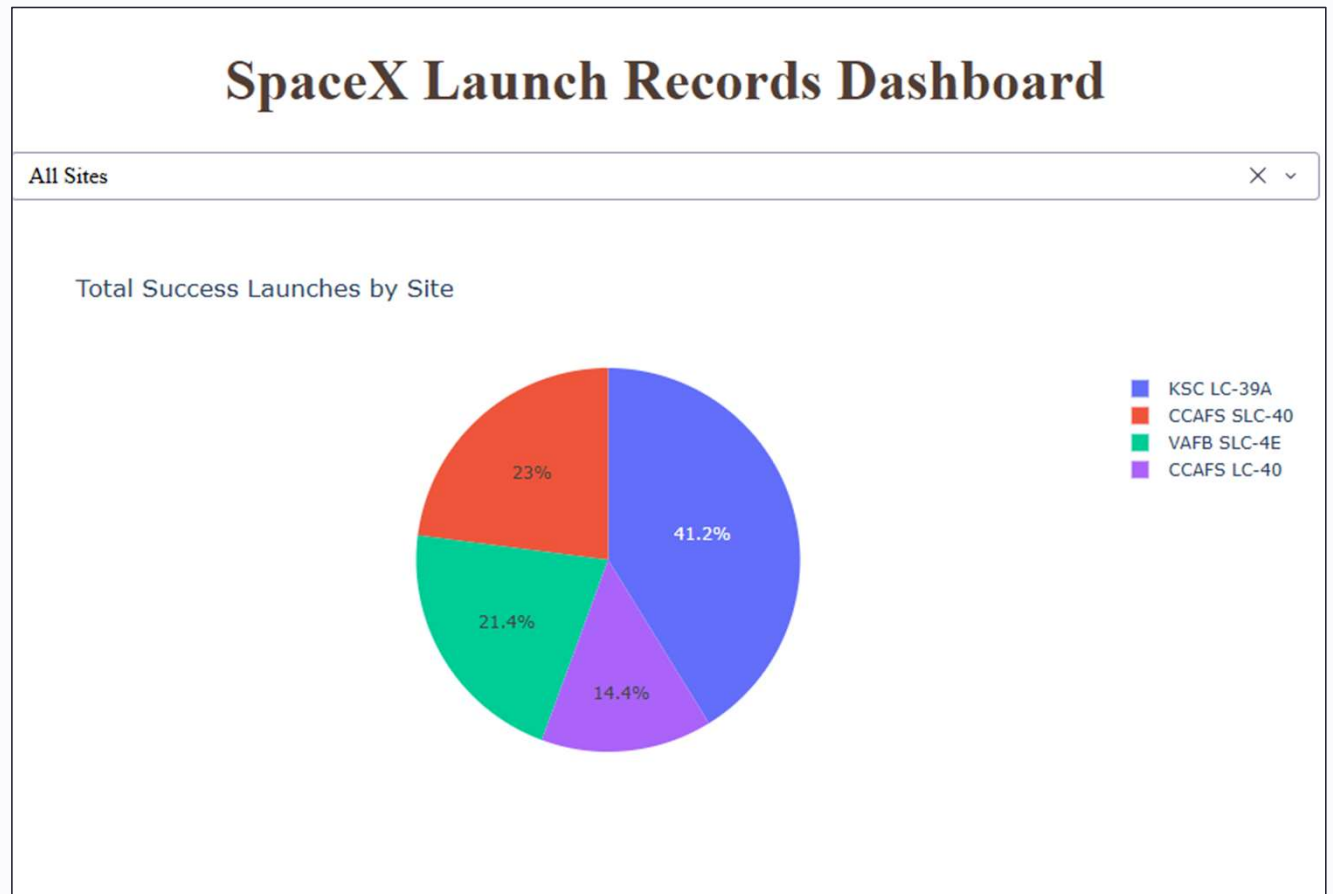
Section 4

Build a Dashboard with Plotly Dash



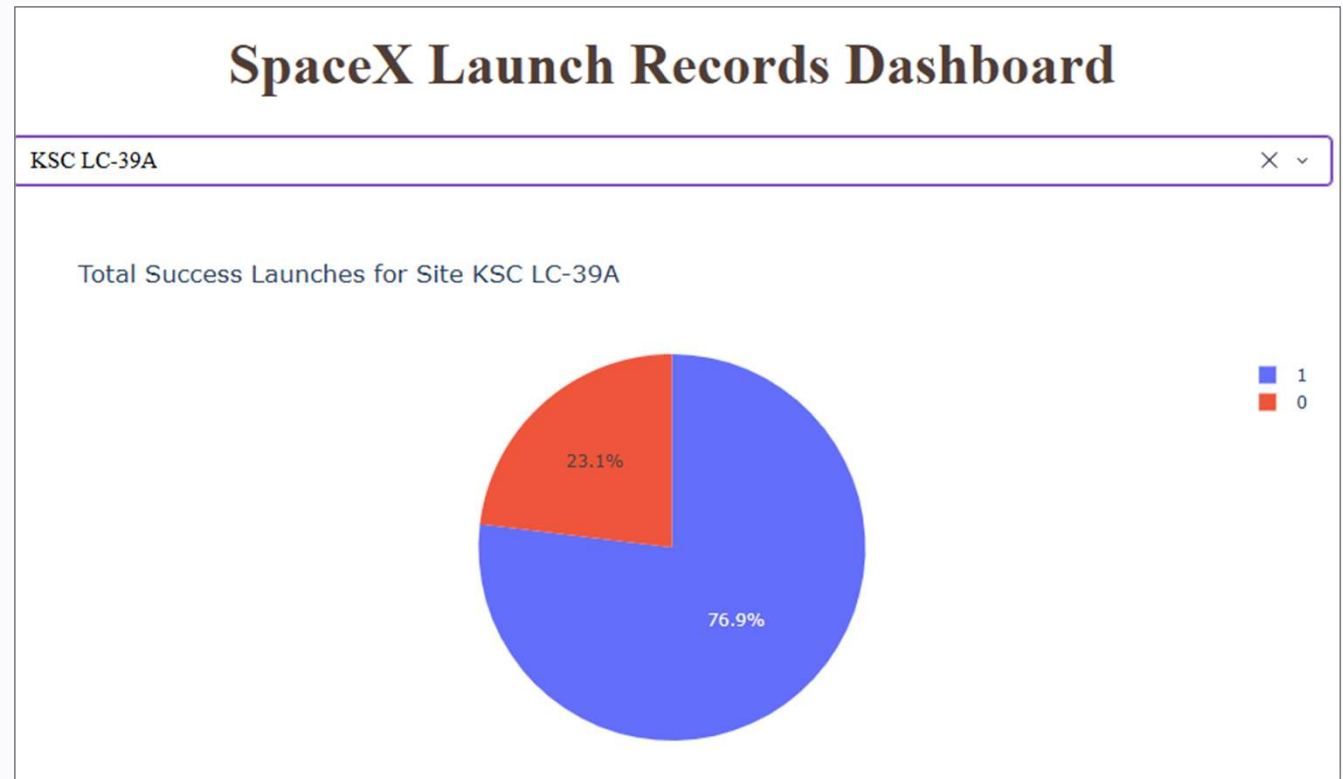
DASHBOARD – Pie chart showing the total success launches by site in percentage

The launch site KSC LC-39A has the most successful launches 41.2%.



DASHBOARD – Pie chart for the launch site with highest launch success ratio

KSC LC-39A has landed
successfully in 76.9% of all launches

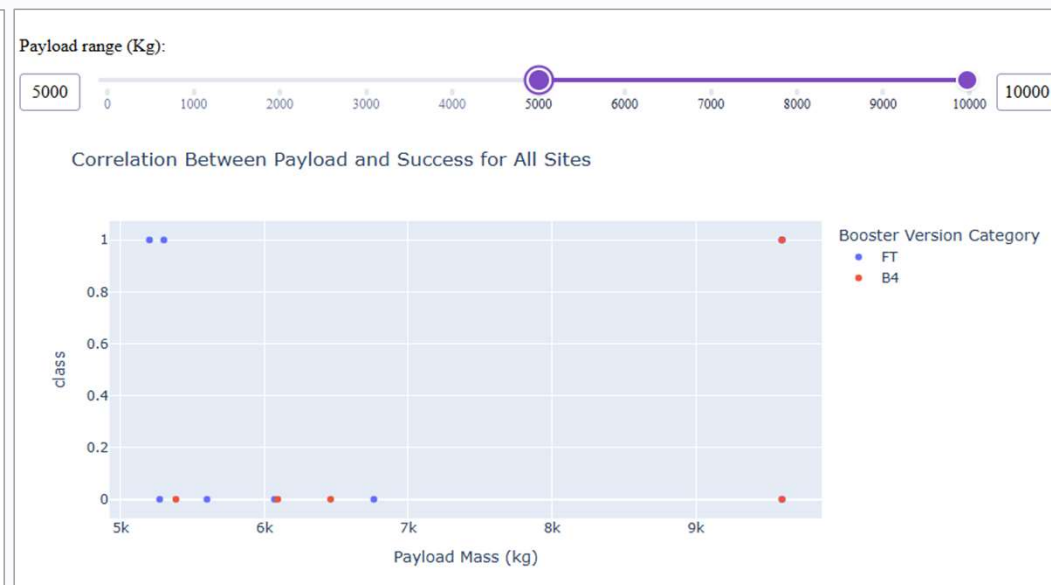


DASHBOARD – Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider

Low Payload



High Payload



- There are much more booster landed for low payload mass (0:5000)kg than for high payload mass (5000:10000)kg

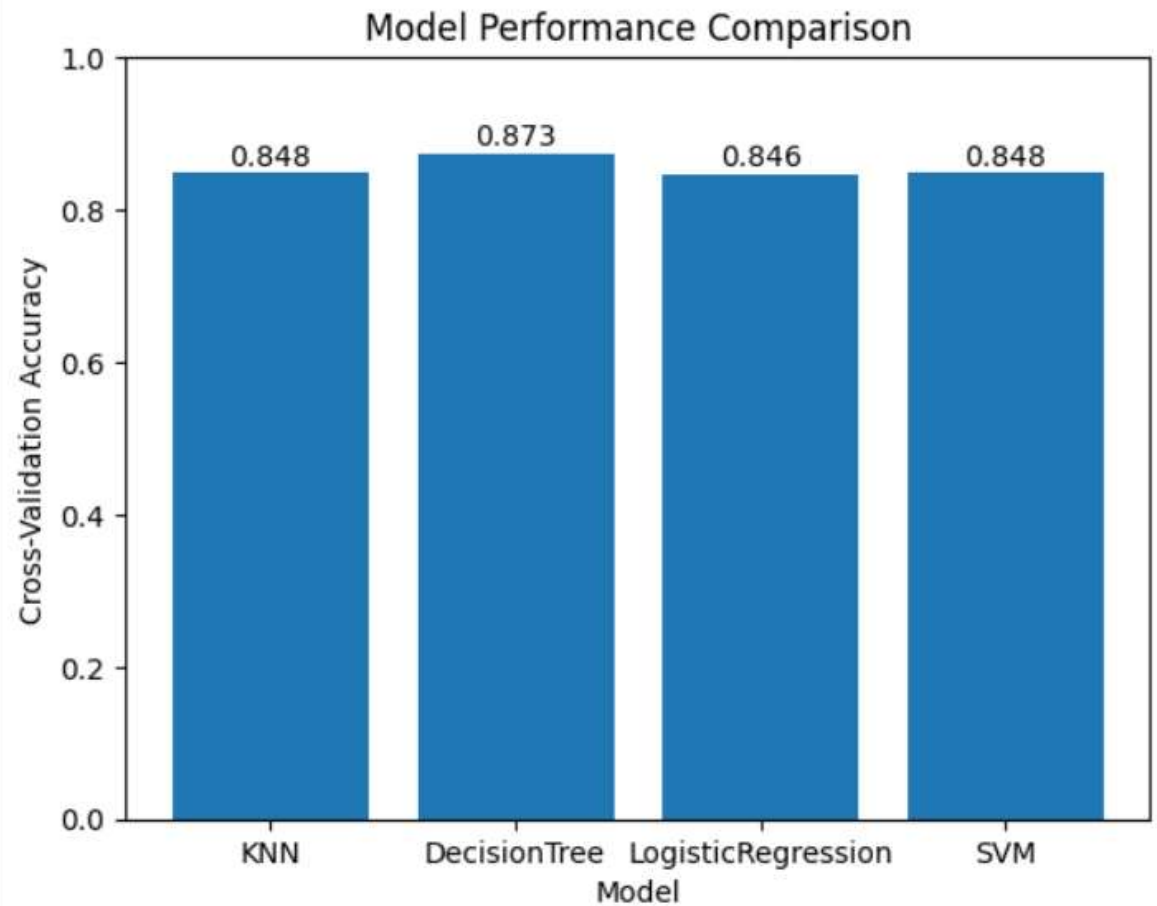
Section 5

Predictive Analysis (Classification)



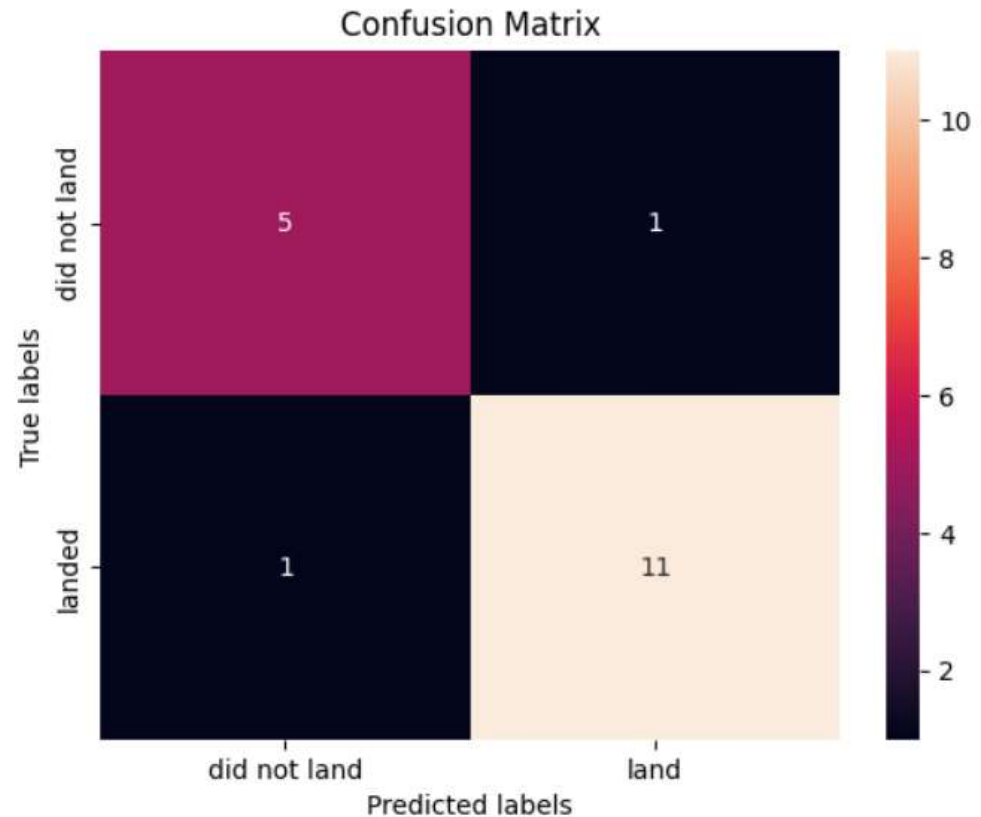
Classification Accuracy

- Best Algorithm is DecisionTree with a score of 0.8732142857142857



Confusion Matrix of the Decision Tree model

- True Negatives (TN) = 5: 5 instances that actually did not land were correctly predicted as “did not land.”
- False Positives (FP) = 1: 1 instance that did not land was incorrectly predicted as “land.”
- “False Negatives (FN) = 1: 1 instance that actually landed was incorrectly predicted as “did not land.”
- “True Positives (TP) = 11: 11 instances that actually landed were correctly predicted as “land.”
- The model therefore correctly classified 16 out of 18 observations. The low number of false positives and false negatives indicates that the model is relatively effective at distinguishing between the two classes, “land” and “did not land.”



Conclusions

- The Decision Tree Classifier achieved the best performance among the machine learning models tested on this dataset, making it the most suitable model
- Lower payload weights generally showed higher success rates compared with heavier payloads, suggesting that payload weight may influence landed success.
- The analysis indicates that SpaceX's landed success rate has improved over time.
- KSC LC-39A recorded the highest number of successful landings among the launch sites analyzed, making it one of the most successful launch locations in the dataset.
- The GEO, HEO, SSO, and ES-L1 orbits showed the highest success rates, indicating that launches targeting these orbits have historically had better outcomes in the analyzed dataset.
- Overall, the analysis demonstrates that **launch site, payload weight, orbit, and time** are important factors associated with SpaceX launch success, while the Decision Tree model provides a useful approach for predicting future outcomes.